

AN AUTOMATED STOCHASTIC VOLATILITY MODEL PARAMETER CALIBRATION ROUTINE WITH OPTIONAL JUMPS

RADU BRICIU

ABSTRACT. We specify a routine that calibrates the five canonical Heston [1993] parameters $\Phi = (\theta, \kappa, \eta, \rho, v_0)$ to a day of CBOE S&P 500 (SPX) intraday option trades. The routine performs *one* calibration per trading day over a pooled, moneyness-normalised implied-volatility surface, and is framed throughout as a box-constrained nonlinear least-squares problem solved from multiple seeds. We make explicit every modelling choice: the volume-weighted reference spot S_{ref} , the moneyness-normalised and grid-snapped strike K^* , the surface-coverage requirements, the configurable in-engine objective (defaulting to implied-volatility “IV-space” residuals), the IV-space restart selection and acceptance gate, and the boundary-pegging rejection rule. The same routine optionally fits the Bates [1996] extension, which adds Merton lognormal jumps to the Heston dynamics, selected by a run-time flag and calibrated over the identical surface and gate. We keep Heston as the default throughout and present Bates as the eight-parameter generalisation it specialises from.

CONTENTS

1.	The model	1
1.1.	Basis	1
1.2.	Jumps	1
1.3.	Pricing operators	1
2.	Data and per-day notation	3
3.	Handling intraday spot movement	3
4.	Surface construction	4
5.	The optimization problem	5
5.1.	Residuals	5
5.2.	Multi-start local optimization	6
6.	Acceptance gate	6
7.	Output and diagnostics	7
8.	Constants	8
9.	Avenues for improvement	9
	References	9
	Appendix A. Example option prices	10
	Appendix B. Example fits for the Heston model	12
	Appendix C. Example fits for the Bates model	14
	Appendix D. Algorithm	16

1. The model

1.1. Basis. Under the risk-neutral measure, with constant risk-free rate r and continuous dividend yield g , the log-price $X_t = \ln S_t$ and instantaneous variance v_t of the SPX follow the Heston [1993] dynamics

$$dX_t = (r - g - \frac{1}{2}v_t) dt + \sqrt{v_t} (\rho dW_t + \sqrt{1 - \rho^2} dB_t), \quad (1)$$

$$dv_t = \kappa(\theta - v_t) dt + \eta\sqrt{v_t} dW_t, \quad (2)$$

where W, B are independent Brownian motions, so $\text{corr}(dX_t, dv_t) = \rho dt$. The parameter vector is

$$\Phi = (\theta, \kappa, \eta, \rho, v_0) \in \mathbb{R}^5, \quad (3)$$

with $\theta > 0$ the long-run variance, $\kappa > 0$ the mean-reversion speed, $\eta > 0$ the volatility of variance (QuantLib's `sigma`), $\rho \in [-1, 1]$ the leverage correlation, and $v_0 > 0$ the initial variance. The Feller quantity [Feller, 1951]

$$\mathcal{F}(\Phi) = 2\kappa\theta - \eta^2 \quad (4)$$

is reported per fit; $\mathcal{F}(\Phi) \geq 0$ guarantees $v_t > 0$ almost surely. It is not imposed as a constraint (short-tenor SPX fits routinely violate it without being otherwise unreasonable; see §6).

1.2. Jumps. The Bates [1996] model extends Heston with Merton [1976] log-normal jumps in the log-price. The variance dynamics (2) are unchanged. The level equation (1) gains a compound-Poisson jump term:

$$dX_t = (r - g - \frac{1}{2}v_t - \lambda\xi) dt + \sqrt{v_t} (\rho dW_t + \sqrt{1 - \rho^2} dB_t) + dJ_t. \quad (5)$$

Here $J_t = \sum_{n=1}^{N_t} Y_n$ is a compound-Poisson process: N_t is a Poisson process of intensity $\lambda \geq 0$ (expected jumps per year), independent of W and B , and the log-jump sizes $Y_n \sim \mathcal{N}(\nu, \delta^2)$ are i.i.d. The compensator

$$\xi = \mathbb{E}[e^Y - 1] = e^{\nu + \delta^2/2} - 1 \quad (6)$$

offsets the jump drift so the discounted price stays a martingale, leaving the level drift $r - g$ in expectation. The augmented parameter vector is

$$\Theta = (\theta, \kappa, \eta, \rho, v_0, \lambda, \nu, \delta) \in \mathbb{R}^8, \quad (7)$$

the five Heston parameters of (1)–(2) plus jump intensity $\lambda \geq 0$, mean log-jump ν , and log-jump volatility $\delta > 0$. At $\lambda = 0$ the jump term vanishes and (5) collapses to the Heston level (1), so Heston is the $\lambda = 0$ face of the Bates parameter space. The model is selected at run time by a flag; the Heston fit is the default. The Feller quantity (4) is built from the diffusive parameters alone, so jumps do not enter it; it is reported under both models.

1.3. Pricing operators. Fix a valuation with maturity $\tau > 0$ (in years), strike K , and type $w \in \{+1, -1\}$ (+1 call, -1 put). Write the forward and discount factor under flat curves as

$$F(S, \tau) = S e^{(r-g)\tau}, \quad D(\tau) = e^{-r\tau}. \quad (8)$$

Black price [Black, 1976, Black and Scholes, 1973]. With total standard deviation $\Sigma = \sigma\sqrt{\tau}$,

$$C_B(\sigma; S, K, \tau, w) = w D(\tau) \left[F \mathcal{N}(wd_+) - K \mathcal{N}(wd_-) \right], \quad (9)$$

$$d_{\pm} = \frac{\ln(F/K)}{\Sigma} \pm \frac{1}{2}\Sigma,$$

where $\mathcal{N}$ is the standard normal CDF and $F = F(S, \tau)$.

Heston price. The Heston model price $C_H(\Phi; S, K, \tau, w)$ is the semi-analytic value

$$C_H(\Phi; S, K, \tau, +1) = S e^{-g\tau} P_1 - K e^{-r\tau} P_2, \quad (10)$$

$$C_H(\cdot, -1) = C_H(\cdot, +1) - S e^{-g\tau} + K e^{-r\tau},$$

with the in-the-money probabilities, for $j \in \{1, 2\}$,

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[\frac{e^{-iu \ln K} f_j(u; \Phi, \tau, \ln F)}{iu} \right] du, \quad (11)$$

and characteristic functions $f_j = \exp(C_j + D_j v_0 + iu \ln F)$ given by the standard coefficients

$$C_j = \frac{\kappa\theta}{\eta^2} \left[(b_j - \rho\eta iu + d_j)\tau - 2 \ln \frac{1 - \gamma_j e^{d_j\tau}}{1 - \gamma_j} \right], \quad (12)$$

$$D_j = \frac{b_j - \rho\eta iu + d_j}{\eta^2} \frac{1 - e^{d_j\tau}}{1 - \gamma_j e^{d_j\tau}},$$

$$d_j = \sqrt{(\rho\eta iu - b_j)^2 - \eta^2(2\mu_j iu - u^2)}, \quad \gamma_j = \frac{b_j - \rho\eta iu + d_j}{b_j - \rho\eta iu - d_j}, \quad (13)$$

where $\mu_1 = \frac{1}{2}$, $\mu_2 = -\frac{1}{2}$, $b_1 = \kappa - \rho\eta$, $b_2 = \kappa$. In code this integral is evaluated by the QuantLib [Ballabio and the QuantLib contributors] **AnalyticHestonEngine** (which uses a branch-stable formulation of (11)); equations (10)–(11) define the operator it realises.

Bates price. The Bates price $C_{\text{Bates}}(\Theta; S, K, \tau, w)$ has the same form as the Heston price (10)–(11), with each characteristic function f_j multiplied by the Merton lognormal jump factor of §1.2:

$$f_j^B(u) = f_j(u) e^{\tau \psi_j(u)}, \quad \psi_j(u) = \lambda \left(e^{iu_j \nu - \frac{1}{2} \delta^2 u_j^2} - 1 \right) - iu_j \lambda \xi, \quad (14)$$

with $u_2 = u$, $u_1 = u - i$ (the same measure shift that distinguishes f_1 from f_2), and $\xi = e^{\nu + \delta^2/2} - 1$ the compensator (6). The exponent ψ_j is the Merton log-jump cumulant, compensated by $-iu_j \lambda \xi$ so the drift correction of (5) carries through. At $\lambda = 0$ the factor is 1 and C_{Bates} reduces to C_H . In code the integral is evaluated by the QuantLib **BatesEngine**, the jump-augmented counterpart of the **AnalyticHestonEngine**.

Implied volatility. The Black-implied volatility of a price C is the inverse of (9) in its first argument,

$$\mathcal{I}(C; S, K, \tau, w) = \{ \sigma > 0 : C_B(\sigma; S, K, \tau, w) = C \}, \quad (15)$$

computed numerically (QuantLib **blackFormulaImpliedStdDev** on the forward (8), returning Σ , then $\sigma = \Sigma/\sqrt{\tau}$). This operator defines the IV-space residuals of §5.

2. Data and per-day notation

The pipeline ingests CBOE trade-level files, one per trading day d , restricted upstream to out-of-the-money (OTM) quotes. A day is a set of trade records indexed by i , each carrying

$$(t_i, K_i, w_i, q_i, P_i, \sigma_i, S_i, \tau_i), \quad (16)$$

being respectively the quote timestamp, strike, type $w_i \in \{+1, -1\}$, traded size, trade price, trade-implied volatility, the underlying level *at trade time* t_i , and the days to maturity. Maturities enter the engine as τ_i in years under Actual/365, i.e. $T_i = \tau_i/365$.

The day is rate-stamped with $r = r(d)$ and $g = g(d)$, the most recent risk-free and dividend rates on or before d (a flat-forward curve for each; under flat curves the evaluation date is immaterial to T_i , so it is set to the QuantLib “today”).

Pre-filter. We keep only economically usable, Heston-appropriate quotes. Write the ratio moneyness $\varrho_i = e^{-|\ln(K_i/S_i)|} \in (0, 1]$ (it equals 1 at the money and falls as the strike moves out). The kept set is

$$\mathcal{D} = \{i : \sigma_i > 0, \tau_{\min} \leq \tau_i \leq \tau_{\max}, \underline{m} < \varrho_i < \bar{m}\}. \quad (17)$$

The maturity floor $\tau_{\min}$ (Table 1) drops ultra-short maturities, which the diffusive Heston smile fits poorly and which otherwise drive η, κ to Feller-violating extremes that pollute the pooled fit; the ceiling $\tau_{\max}$ drops thin, stale long tenors. The cutoff $\bar{m}$ removes the near-ATM rows, and the floor $\underline{m}$ removes the deep-OTM lottery-ticket tail ($\varrho_i < \underline{m}$, roughly $|\ln(K_i/S_i)| > 0.51$), whose extreme prices peg the fit to its bounds. Flooring that tail recovers acceptance and tightens the fit while keeping the full tradeable wing.

3. Handling intraday spot movement

A Heston calibration is anchored to a *single* spot S , yet the trades in $\mathcal{D}$ occur across an intraday range of underlying levels $\{S_i\}$. We reconcile the two with one reference spot and a moneyness re-striking.

Reference spot. Define the volume-weighted reference spot

$$S_{\text{ref}} = \frac{\sum_{i \in \mathcal{D}} q_i S_i}{\sum_{i \in \mathcal{D}} q_i}. \quad (18)$$

The intraday range and its flag are

$$R = \frac{\max_i S_i}{\min_i S_i} - 1, \quad \text{HIGH_MOVE} = \mathbf{1}\{R > R_{\max}\}. \quad (19)$$

A flagged day ($R > R_{\max}$) is still calibrated and written, but the normalisation below is recorded as strained (its S_{ref} should be treated with suspicion).

Moneyness normalisation and grid snap. Each trade retains its moneyness at trade time and is re-struck to the reference spot, then snapped to the SPX strike grid of width Δ :

$$m_i = \frac{K_i}{S_i}, \quad K_i^* = \Delta \cdot \text{round}\left(\frac{m_i S_{\text{ref}}}{\Delta}\right). \quad (20)$$

The rationale is sticky-moneyness: over a session the implied-volatility smile is approximately stationary in K/S rather than in the absolute strike. Re-striking by (20) therefore re-centres the whole day onto S_{ref} so that trades executed at

different S_i but the *same* moneyness share a common surface column. The grid snap collapses the continuum of K_i^* values (one per trade) back onto a clean strike lattice. Pricing diagnostics (§7) nonetheless use each contract’s *original* (S_i, K_i) , since the calibrated parameters are spot-independent.

4. Surface construction

From the normalised trades we build a partial $K^* \times \tau$ implied-volatility surface, selecting the most liquid quotes nearest the money. The trades entering this stage are already the pre-filtered set $\mathcal{D}$, so every candidate strike sits in the OTM moneyness band $\underline{m} < \varrho_i < \bar{m}$ of (17). That band fixes the reach of each wing: the cutoff $\bar{m}$ sets the inner edge, the strikes closest to the money, and the floor $\underline{m}$ sets the outer edge, where the deep-OTM tail is dropped. “Nearest the money” below therefore means just inside $\bar{m}$, not at the money itself.

Maturity selection. A maturity enters the surface only if it carries both wings. For each distinct maturity τ present in $\mathcal{D}$, let $n_c(\tau)$ and $n_p(\tau)$ be the number of distinct K^* on its call and put wings, and let $V(\tau) = \sum_{i \in \mathcal{D}: \tau_i = \tau} q_i$ be its traded volume. Call τ *admissible* when each wing carries at least $\underline{n}_k$ distinct strikes. Keep the $\bar{N}_t$ highest-volume admissible maturities,

$$\mathcal{T} = \text{top-}\bar{N}_t \text{ by } V(\tau) \text{ among } \{\tau : n_c(\tau) \geq \underline{n}_k, n_p(\tau) \geq \underline{n}_k\}. \quad (21)$$

The wing test precedes the volume cap. A maturity with a one-sided or single-strike wing is discarded before the cap, rather than consuming one of its $\bar{N}_t$ slots. The kept set falls below $\bar{N}_t$ only when fewer than $\bar{N}_t$ maturities are admissible.

Strike selection (per maturity, per wing). Within each kept maturity $\tau \in \mathcal{T}$, separate calls ($w_i = +1$, OTM with $K_i^* \gtrsim S_{\text{ref}}$) from puts ($w_i = -1$, OTM with $K_i^* \lesssim S_{\text{ref}}$) and keep the $\bar{n}_k$ nearest-the-money distinct strikes on each wing:

$$\mathcal{K}_\tau = \underbrace{\{\bar{n}_k \text{ largest } K^* \text{ among puts at } \tau\}}_{\text{put wing}} \cup \underbrace{\{\bar{n}_k \text{ smallest } K^* \text{ among calls at } \tau\}}_{\text{call wing}}. \quad (22)$$

The $\bar{n}_k$ cap binds only where a wing carries more than $\bar{n}_k$ distinct strikes inside the band; otherwise the wing’s outer edge is set by the floor $\underline{m}$ alone.

Pivot to a surface. For each retained cell (k, τ) with $k \in \mathcal{K}_\tau$, set the surface entry to the implied volatility of the *highest-volume* trade in that cell, the one with the largest traded size,

$$\hat{\sigma}_{k,\tau} = \sigma_{i^*}, \quad i^* = \arg \max_{i: K_i^* = k, \tau_i = \tau} q_i, \quad (23)$$

and leave it missing otherwise. When several trades tie at the maximum size the pick is arbitrary. Let

$$\mathcal{N} = \{(k, \tau) : \hat{\sigma}_{k,\tau} \text{ defined}\} \quad (24)$$

be the set of populated cells; these become the calibration targets (helpers).

Coverage requirement. To keep the five-parameter fit identified we require a genuinely two-dimensional surface:

$$|\{\tau : (\cdot, \tau) \in \mathcal{N}\}| \geq M_t, \quad |\{k : (k, \cdot) \in \mathcal{N}\}| \geq M_k, \quad |\mathcal{N}| \geq M_{\text{cell}}. \quad (25)$$

A day failing (25) (or with no usable maturities) is dropped: it contributes no calibration and clears its stale diagnostics.

5. The optimization problem

The routine calibrates one parameter vector per day. Under the default Heston model this is Φ of (3) with $\dim \Phi = 5$; under Bates it is Θ of (7) with $\dim \Theta = 8$. The two are nested. They share their first $\dim \Phi$ coordinates, the diffusive parameters $(\theta, \kappa, \eta, \rho, v_0)$, and Θ appends the jump triple (λ, ν, δ) . Setting $\lambda = 0$ collapses the jump term (5) and recovers Φ , so Heston is the $\lambda = 0$ face of Bates. We write Ψ for the active vector, either Φ or Θ , and $C^{\text{mod}}(\Psi; \cdot)$ for the active model price: the Heston price C_H (10) when $\Psi = \Phi$, and the Bates price C_{Bates} (14) when $\Psi = \Theta$. Everything below is written in Ψ and holds for either model. The only place the model enters is the box dimension $\dim \Psi$ (26) and, through it, the bounds tables.

Let $\mathcal{N}$ index the targets with market data $\{(\hat{\sigma}_n, K_n^*, T_n, w_n)\}_{n \in \mathcal{N}}$, all priced at the single spot S_{ref} . Define the feasible box

$$\mathcal{B} = \prod_{j=1}^{\dim \Psi} [\ell_j, h_j] \subset \mathbb{R}^{\dim \Psi}, \quad (26)$$

with (ℓ, h) in the `model.params()` order of Table 2: the five Heston coordinates, extended by the three jump bounds when $\Psi = \Theta$. The λ floor is exactly 0, so the Bates box contains the Heston model on its $\lambda = 0$ face.

5.1. Residuals. Two residual systems are defined: an IV-space system and a price-space system. The in-engine objective is configurable. It defaults to the IV-space system (30) below, and can instead be set to the price-space system (27). The choice changes only what each restart minimises. Selection (34) and the acceptance gate (36) always run on the IV-space error J_{iv} , so under the default objective the optimizer minimises the very error that then selects and gates.

Price-space (alternative in-engine objective). For each target the market value is the Black price at the quoted vol, $C_n^{\text{mkt}} = C_B(\hat{\sigma}_n; S_{\text{ref}}, K_n^*, T_n, w_n)$, and the relative-price residual is

$$\varepsilon_n^{\text{px}}(\Psi) = \frac{C^{\text{mod}}(\Psi; S_{\text{ref}}, K_n^*, T_n, w_n) - C_n^{\text{mkt}}}{C_n^{\text{mkt}}}. \quad (27)$$

The associated objective is the root-mean-square

$$J_{\text{px}}(\Psi) = \left(\frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{N}} \varepsilon_n^{\text{px}}(\Psi)^2 \right)^{1/2}. \quad (28)$$

The per-target residual $\varepsilon_n^{\text{px}}$ is the (relative-price) error each QuantLib helper exposes. Levenberg–Marquardt [Marquardt, 1963] is a sum-of-squares method, so it drives the squared residual norm $\sum_n (\varepsilon_n^{\text{px}})^2 = |\mathcal{N}| J_{\text{px}}(\Psi)^2$ to a local minimum (equivalently J_{px} , with which it shares a minimiser).

IV-space (default in-engine objective; selection and gate). Invert each model price back to a Black vol via (15) and compare to the market vol:

$$\begin{aligned} \sigma_n^{\text{mod}}(\Psi) &= \mathcal{I}(C^{\text{mod}}(\Psi; S_{\text{ref}}, K_n^*, T_n, w_n); S_{\text{ref}}, K_n^*, T_n, w_n), \\ \varepsilon_n^{\text{iv}}(\Psi) &= \sigma_n^{\text{mod}}(\Psi) - \hat{\sigma}_n, \end{aligned} \quad (29)$$

with objective

$$J_{\text{iv}}(\Psi) = \left(\frac{1}{|\mathcal{N}'|} \sum_{n \in \mathcal{N}'} \varepsilon_n^{\text{iv}}(\Psi)^2 \right)^{1/2}, \quad \mathcal{N}' = \{n : \sigma_n^{\text{mod}}(\Psi) \text{ inverts finitely}\}. \quad (30)$$

J_{iv} is measured in volatility points; it is the natural surface-fit metric and, unlike J_{px} , is not inflated by cheap deep-OTM wings (a unit price error on a near-worthless contract is a large *relative* price error but a small vol error).

5.2. Multi-start local optimization. A single start into a non-convex problem is fragile [Mikhailov and Nögel, 2003], so we restart from a small, data-seeded set. Let the surface variance level be

$$\hat{v} = \text{clip}\left(\left(\text{median}_{(k,\tau) \in \mathcal{N}} \hat{\sigma}_{k,\tau}\right)^2, [10^{-3}, 0.25]\right), \quad (31)$$

and define the seed set $\mathcal{S} = \{s_1, \dots, s_6\} \subset \mathcal{B}$ of $(v_0, \kappa, \theta, \eta, \rho)$ tuples

$$\mathcal{S} = \begin{cases} (\hat{v}, 1.0, \hat{v}, 0.50, -0.70), & (\hat{v}, 3.0, \hat{v}, 1.00, -0.50), \\ (\hat{v}, 0.5, \hat{v}, 0.30, -0.90), & (\hat{v}, 5.0, \hat{v}, 0.80, -0.60), \\ (0.8\hat{v}, 2.0, 1.2\hat{v}, 0.60, -0.75), & (\hat{v}, 8.0, \hat{v}, 1.20, -0.40). \end{cases} \quad (32)$$

Under Bates each seed is extended by one fixed jump triple $(\lambda, \nu, \delta) = (0.1, -0.1, 0.1)$, so every restart starts from almost no jumps and grows them only if they tighten the fit; the restart count is unchanged. For each seed $s \in \mathcal{S}$ we solve the box-constrained nonlinear least-squares problem

$$\hat{\Psi}_s = \arg \min_{\Psi \in \mathcal{B}} \sum_{n \in \mathcal{N}} \varepsilon_n^{\text{obj}}(\Psi)^2 \quad (\text{LM, started at } \Psi^{(0)} = s), \quad (33)$$

using Levenberg–Marquardt under the box constraint $\mathcal{B}$ (`QuantLib NonhomogeneousBoundaryConstraint`). The per-target residual $\varepsilon_n^{\text{obj}}$ is the configured in-engine objective: by default the IV-space residual $\varepsilon_n^{\text{iv}}$ (29), optionally the price-space residual $\varepsilon_n^{\text{px}}$ (27). The fit retained for the day is the restart with the smallest *IV-space* error,

$$\Psi^* = \arg \min_{s \in \mathcal{S}} J_{\text{iv}}(\hat{\Psi}_s). \quad (34)$$

Ranking by J_{iv} (not J_{px}) removes the dependence on any single arbitrary guess and selects the economically best surface fit.

6. Acceptance gate

Ψ^* is accepted only if it fits the surface tightly *and* lies in the interior of the box. Let the boundary-pegging predicate, with tolerance fraction b_{tol} , be

$$\text{PEGGED}(\Psi) = \bigvee_{j=1}^{\dim \Phi} \left[(\Psi_j - \ell_j) < b_{\text{tol}} (h_j - \ell_j) \vee (h_j - \Psi_j) < b_{\text{tol}} (h_j - \ell_j) \right]. \quad (35)$$

The disjunction runs over the $\dim \Phi$ diffusive coordinates for both models. Under Bates the three jump coordinates (ν, δ, λ) are exempt from the predicate: $\lambda \rightarrow 0$ is a legitimate collapse to Heston rather than the optimizer hitting a wall, and when $\lambda \approx 0$ the jump shape (ν, δ) is unidentified and may rest on a bound without meaning. The jump triple is still reported. The Feller quantity (4)

likewise stays the diffusive quantity under Bates and never gates. The day is accepted iff

$$\boxed{J_{\text{iv}}(\Psi^*) \leq \varepsilon_{\text{iv}} \quad \text{and} \quad \neg \text{PEGGED}(\Psi^*)}. \quad (36)$$

The first clause is the IV-space gate ($\varepsilon_{\text{iv}} \approx 2$ vol points). The second rejects boundary fits: a parameter resting on a bound is the optimizer hitting a wall, not an identified value, so it is treated as a non-fit regardless of J_{iv} . The Feller quantity (4) is reported but never gates, since short-tenor SPX fits can violate it while remaining otherwise sound. A rejected or too-thin day yields no parameter row.

Why an IV-space gate. An earlier version gated on J_{px} directly. On a pooled day rich in deep-OTM wings, J_{px} is dominated by those cheap contracts and rejects fits that are excellent in vol points. Replacing the gate metric with J_{iv} aligns the acceptance criterion with the units the surface is quoted in; under it the genuine pooled-surface fit is ≈ 0.8 – 1.4 vol points, and the residual rejections are boundary-pegged κ or ρ (a separate identification lever), not fit-quality failures.

7. Output and diagnostics

Every attempted day produces output. Results are routed by model and objective into one directory, `results/⟨model⟩/calibrations/⟨objective⟩/`, so a Heston run never shares a file with a Bates run, nor an IV-space run with a price-space one. That directory holds three artefacts: a parameter table of accepted days (`calibrations.csv`), a rejection table of dropped days (`rejections.csv`), and a folder of per-day repricing diagnostics (`calibration_tests/`). The two tables partition the attempted days. An accepted day writes one row to the parameter table and nothing to the rejection table; a rejected day does the reverse. So the accept rate and the rejection breakdown of §?? read straight off them.

Accepted days: the parameter table. An accepted day appends one row, keyed by the date d , to `calibrations.csv`. The row carries the rate-stamped inputs, the fitted vector, and fit and coverage diagnostics:

$$\begin{aligned} & (d; S_{\text{ref}}, r, g; \theta^*, \kappa^*, \rho^*, \eta^*, v_0^*; \mathcal{F}(\Psi^*); \\ & J_{\text{iv}}(\Psi^*), J_{\text{px}}(\Psi^*), |\mathcal{N}|; |\mathcal{T}|, \#\text{strikes}, |\mathcal{N}|, \sum_i q_i; \\ & \min_i S_i, \max_i S_i, R, \text{HIGH_MOVE}; t_{\text{calc}}). \end{aligned} \quad (37)$$

The blocks are: the reference spot (18) and the day’s rates; the fitted parameters Ψ^* with their Feller quantity (4); the two objective values (30) (28) and the helper count $|\mathcal{N}|$; the surface coverage (kept maturities, distinct strikes, populated cells) and the day’s total traded volume; the intraday spot range with its flag (19); and t_{calc} , the latest quote timestamp on the surface. A boolean accept flag (always true in this file) is stored alongside. Under Bates the jump triple $(\lambda^*, \nu^*, \delta^*)$ is inserted after v_0^* and nothing else changes. Because the calibration is once-daily, these rows accumulate into the one file, rewritten in date order at the end of each run.

Rejected days: the rejection table. A day dropped at any stage appends one row, also keyed by d , to `rejections.csv`, recording why:

$$(d; \text{REASON}, \text{DETAIL}; J_{\text{iv}}; |\mathcal{T}|, \#\text{strikes}, |\mathcal{N}|). \quad (38)$$

The `REASON` is one of six categories, in pipeline order: `no_trades` (nothing survives the pre-filter (17)), `no_rate` (no r or g on or before d), `thin` (the coverage requirement (25) fails), `pegged` (the fit clears the IV gate but a diffusive parameter is boundary-pegged (35)), `iv_miss` ($J_{iv}(\Psi^*) > \varepsilon_{iv}$), and `no_fit` (no restart returned a finite fit). `DETAIL` is the human-readable string. The IV error and the coverage counts are filled in where the day reached the stage that computes them, and left blank otherwise: a `thin` day has coverage counts but no J_{iv} , a `no_trades` day has neither. The two tables together cover every attempted day, so the pegged-versus-thin-versus-miss split is auditable directly.

Per-day repricing diagnostics. Each accepted day also writes one file to `calibration.tests/`, holding the day’s repriced surface contracts. One representative trade per populated cell (the highest-volume one, matching the pivot (23)) is valued under Ψ^* . The valuation uses each contract’s *original* (S_i, K_i), not (S_{ref}, K^*): the structural parameters are spot-independent, and the honest comparison is against the actual trade. Each row records the original spot and strike, the snapped K^* , the fitted parameters, the market implied volatility σ_i , the Black price (9) at that vol, and the model price (the Heston price (10), or a Bates price column (14) under Bates).

8. Constants

Symbol	Code name	Value	Role
τ_{min}	MIN_DTM	14	minimum days to maturity
τ_{max}	MAX_DTM	730	maximum days to maturity
$\underline{m}$	OTM_MONEYNESS_FLOOR	0.6	minimum OTM ratio moneyness
$\bar{m}$	OTM_MONEYNESS_CUTOFF	0.98	maximum OTM ratio moneyness
R_{max}	MAX_MOVE_PCT	0.03	intraday-range flag threshold
Δ	STRIKE_GRID	5.0	SPX strike lattice for K^* snap
$\bar{N}_t$	MAX_NT	20	maturities kept (by volume)
$\underline{n}_k$	MIN_NK	2	minimum distinct strikes per wing
$\bar{n}_k$	MAX_NK	40	strikes kept per wing
M_t	MIN_MATS	3	minimum maturities in surface
M_k	MIN_STRIKES	5	minimum strikes in surface
M_{cell}	MIN_CELLS	12	minimum populated cells $ \mathcal{N} $
ε_{iv}	IV_RMSE_ACCEPT	0.02	IV-space acceptance gate (vol pts)
b_{tol}	BOUND_TOL	10^{-3}	boundary-pegging tolerance fraction

TABLE 1. Surface, selection, and gate constants.

	θ	κ	η	ρ	v_0		λ	ν	δ
lower ℓ	10^{-4}	10^{-2}	10^{-2}	-0.999	10^{-4}	lower ℓ	0.0	-0.5	10^{-3}
upper h	1.0	20.0	2.0	0.5	1.0	upper h	5.0	0.2	0.5

TABLE 2. Box bounds $\mathcal{B}$ (26), in the `model.params()` order. Left: the five Heston coordinates ($\theta, \kappa, \eta, \rho, v_0$); the ρ upper bound is kept slightly positive (equity leverage implies $\rho < 0$) and η is capped at 2.0 (SPX vol-of-vol ~ 0.3 – 1.2). Right: the Bates jump bounds (λ, ν, δ) appended when $\Psi = \Theta$; the λ floor is exactly 0 (the fit may collapse to pure Heston) and ν allows more negative room since equity jumps skew down.

9. Avenues for improvement

Rough volatility is the natural next paradigm, but a separate one. Empirical realised variance is well described by a fractional process with Hurst exponent $H \approx 0.1$ [Gatheral et al., 2018], far from the $H = \frac{1}{2}$ that the diffusive Heston and Bates dynamics assume. A rough model fits the steep short-maturity skew that single-factor Heston cannot reach without driving η to its cap, so it speaks directly to the degeneracy and Feller findings above. It does not, however, fit the routine of this paper. The dynamics are non-Markovian, so the finite-dimensional box, the seed grid, and the Feller quantity have no direct analogue. There is no QuantLib engine for it. The fastest tractable variant, rough Heston, prices through a fractional Riccati equation [El Euch and Rosenbaum, 2019] that is far heavier per evaluation than the closed-form Heston integral, and rough Bergomi [Bayer et al., 2016] prices by Monte Carlo. Multi-start calibration over thousands of days is then out of reach without a surrogate pricer. There is also a tension with the present surface: the rough skew is sharpest at the shortest maturities, exactly the tenors the pre-filter drops at $\tau_{\min} = 14$ days (17). A fair test would re-admit the short end, build a rough-Heston pricer or a learned surrogate, and run on a pilot window rather than the full sample. We leave it to separate work.

References

- Luigi Ballabio and the QuantLib contributors. *QuantLib: a free/open-source library for quantitative finance*. URL <https://www.quantlib.org>. Version 1.35.
- David S. Bates. Jumps and Stochastic Volatility: Exchange Rate Processes Implicit in Deutsche Mark Options. *The Review of Financial Studies*, 9(1):69–107, 1996. DOI: [10.1093/rfs/9.1.69](https://doi.org/10.1093/rfs/9.1.69).
- Christian Bayer, Peter Friz, and Jim Gatheral. Pricing under rough volatility. *Quantitative Finance*, 16(6):887–904, 2016. DOI: [10.1080/14697688.2015.1099717](https://doi.org/10.1080/14697688.2015.1099717).
- Fischer Black. The pricing of commodity contracts. *Journal of Financial Economics*, 3(1–2):167–179, 1976. DOI: [10.1016/0304-405X\(76\)90024-6](https://doi.org/10.1016/0304-405X(76)90024-6).
- Fischer Black and Myron Scholes. The Pricing of Options and Corporate Liabilities. *Journal of Political Economy*, 81(3):637–654, 1973. DOI: [10.1086/260062](https://doi.org/10.1086/260062).
- Omar El Euch and Mathieu Rosenbaum. The characteristic function of rough Heston models. *Mathematical Finance*, 29(1):3–38, 2019. DOI: [10.1111/mafi.12173](https://doi.org/10.1111/mafi.12173).
- William Feller. Two Singular Diffusion Problems. *Annals of Mathematics*, 54(1):173–182, 1951. DOI: [10.2307/1969318](https://doi.org/10.2307/1969318).
- Jim Gatheral, Thibault Jaisson, and Mathieu Rosenbaum. Volatility is rough. *Quantitative Finance*, 18(6):933–949, 2018. DOI: [10.1080/14697688.2017.1393551](https://doi.org/10.1080/14697688.2017.1393551).
- Steven L. Heston. A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options. *The Review of Financial Studies*, 6(2):327–343, 1993. DOI: [10.1093/rfs/6.2.327](https://doi.org/10.1093/rfs/6.2.327).
- Donald W. Marquardt. An Algorithm for Least-Squares Estimation of Nonlinear Parameters. *Journal of the Society for Industrial and Applied Mathematics*, 11(2):431–441, 1963. DOI: [10.1137/0111030](https://doi.org/10.1137/0111030).
- Robert C. Merton. Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics*, 3(1–2):125–144, 1976. DOI: [10.1016/0304-405X\(76\)90022-2](https://doi.org/10.1016/0304-405X(76)90022-2).
- Sergei Mikhailov and Ulrich Nögel. Heston’s Stochastic Volatility Model: Implementation, Calibration and Some Extensions. *Wilmott Magazine*, pages 74–79, July 2003.

Appendix A. Example option prices

Heston model prices produced by the market¹ calibrated pricing operator:

$$C_H(\Phi^*; S, K, \tau, w) \quad (10) \quad (36).$$

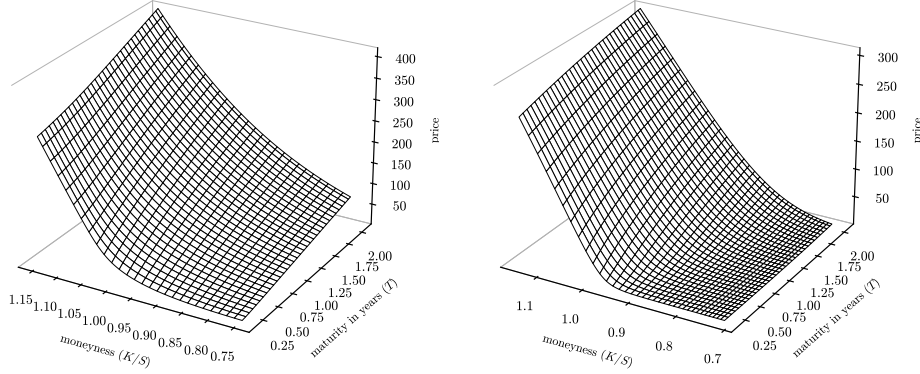


FIGURE 1. Heston option prices for S_{ref} 1957.9267 on October 24, 2014 with $\Phi^* = (0.0538, 3.4097, 1.2288, -0.7475, 0.024)$: puts wing (left) and calls wing (right).

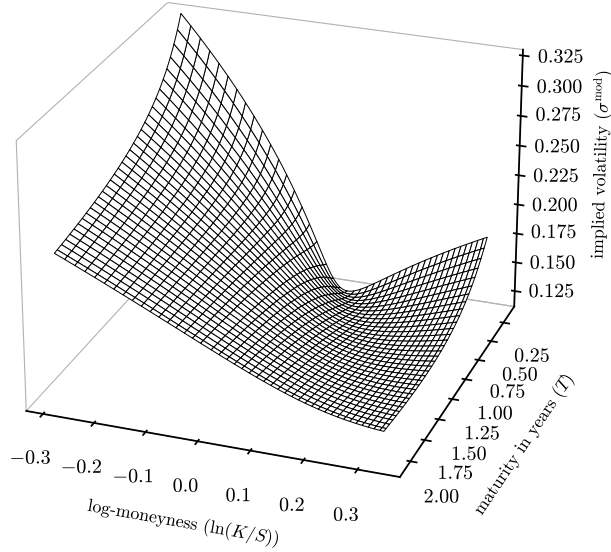


FIGURE 2. Out of the money implied volatilities from Figure 1

¹The fit uses 656 calibration cells across 15 maturities and 140 strikes, backed by 240,081 contracts of traded volume. The reference spot was $S_{\text{ref}} = 1957.9267$, priced under a risk-free rate of 0.09% and a dividend rate of 1.97%. The intraday spot range was 0.97%. Fit quality is 0.54 vol points of implied-volatility RMSE, with a relative-price RMSE of 0.1371. The Feller condition violates at this calibration, with $2\kappa\theta - \eta^2 = -1.1428$.

Bates model prices produced by the market² calibrated pricing operator:

$$C_{\text{Bates}}(\Theta^*; S, K, \tau, w) \quad (14) \quad (36).$$

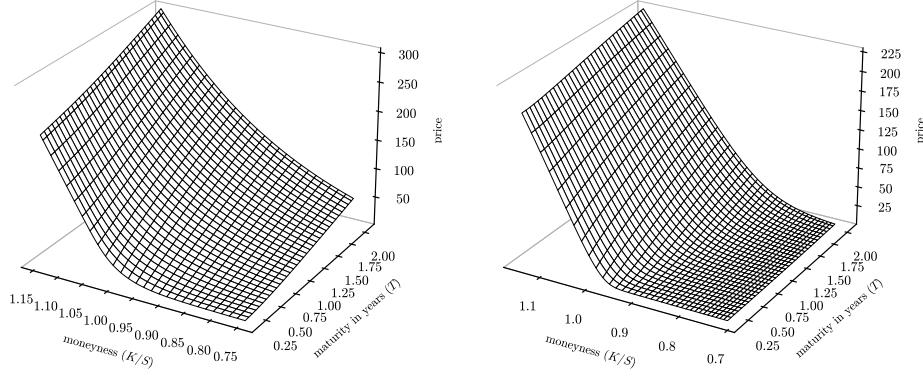


FIGURE 3. Bates option prices for S_{ref} 1500.3753 on January 25, 2013 with $\Theta^* = (0.0615, 0.5521, 0.3767, -0.7631, 0.0103, 0.0723, -0.2699, 0.3295)$: puts wing (left) and calls wing (right).

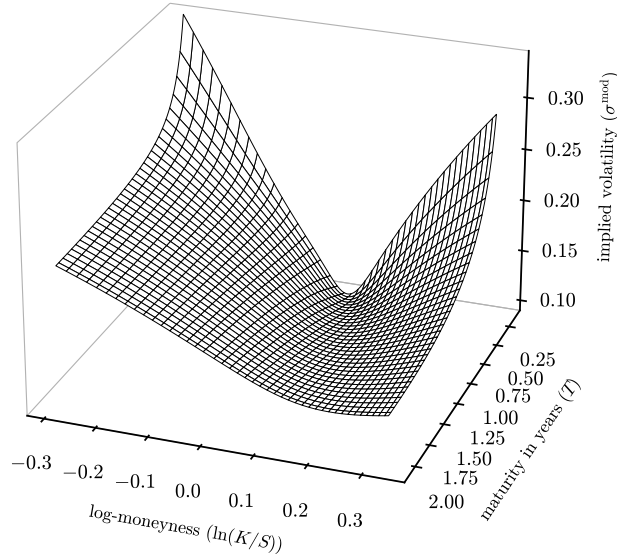


FIGURE 4. *Out of the money* implied volatilities from Figure 3

²The fit uses 418 calibration cells across 13 maturities and 123 strikes, backed by 280,247 contracts of traded volume. The reference spot was $S_{\text{ref}} = 1500.3753$, priced under a risk-free rate of 0.14% and a dividend rate of 2.13%. The intraday spot range was 0.56%. Fit quality is 0.36 vol points of implied-volatility RMSE, with a relative-price RMSE of 0.1217. The Feller condition violates at this calibration, with $2\kappa\theta - \eta^2 = -0.074$.

Appendix B. Example fits for the Heston model

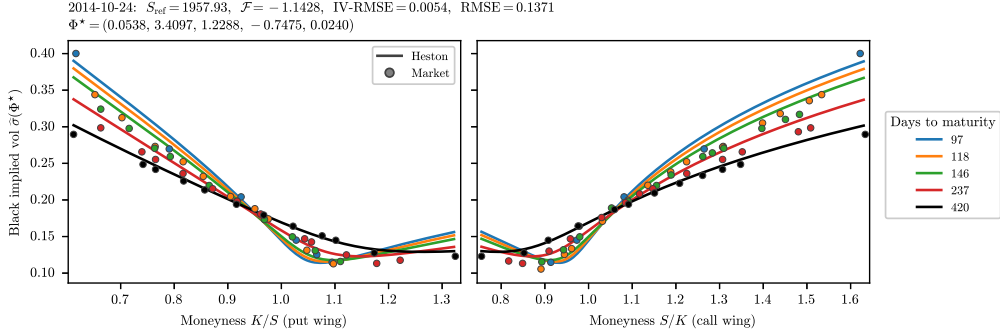


FIGURE 5. Heston implied-volatility smiles for October 24, 2014: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

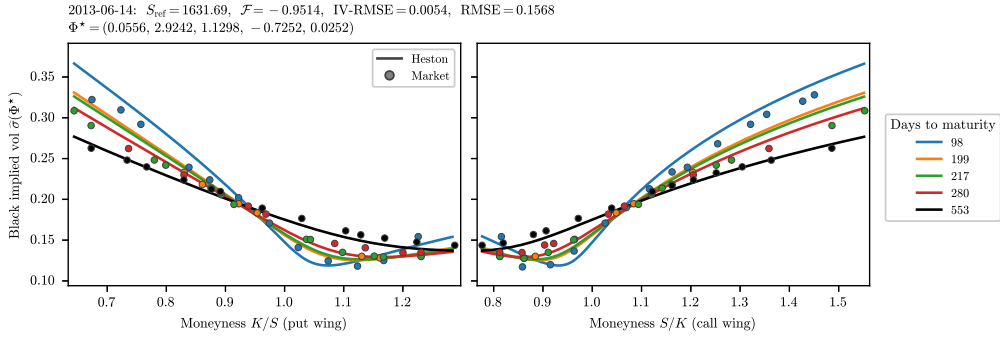


FIGURE 6. Heston implied-volatility smiles for June 14, 2013: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

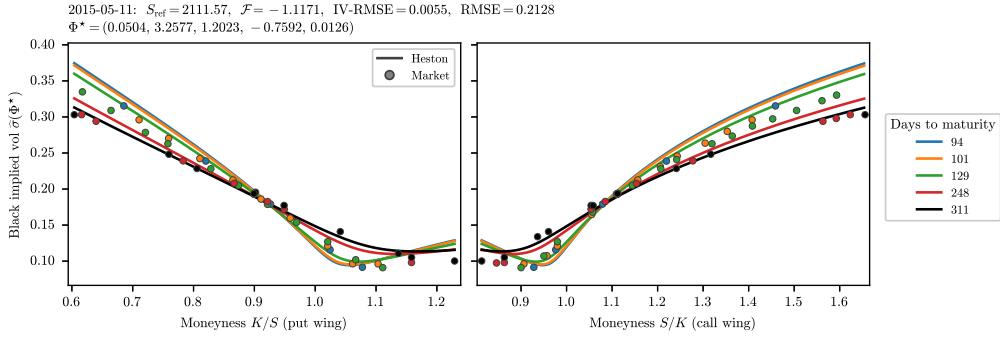


FIGURE 7. Heston implied-volatility smiles for May 11, 2015: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

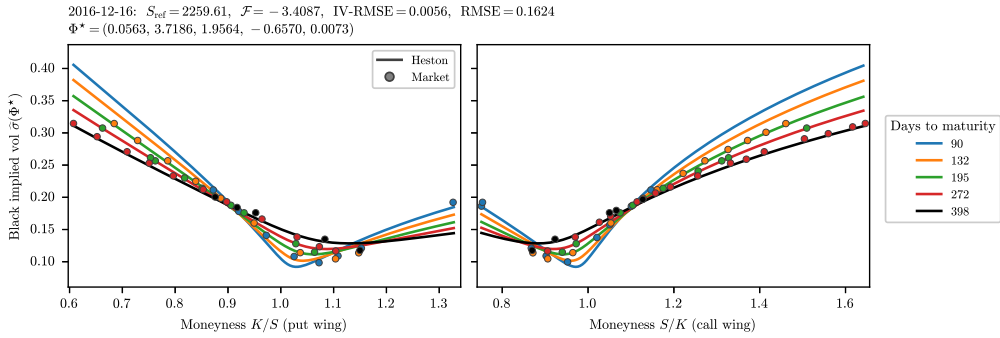


FIGURE 8. Heston implied-volatility smiles for December 16, 2016: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

2016-08-23: $S_{ref} = 2190.37$, $\mathcal{F} = -2.9476$, IV-RMSE = 0.0056, RMSE = 0.1897
 $\Phi^* = (0.0571, 3.5052, 1.8298, -0.7352, 0.0065)$

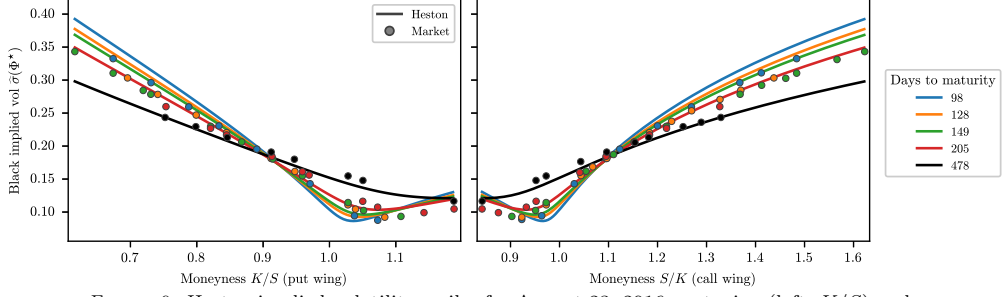


FIGURE 9. Heston implied-volatility smiles for August 23, 2016: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

2017-01-27: $S_{ref} = 2293.84$, $\mathcal{F} = -3.2560$, IV-RMSE = 0.0057, RMSE = 0.1973
 $\Phi^* = (0.0543, 2.9068, 1.8899, -0.6046, 0.0048)$

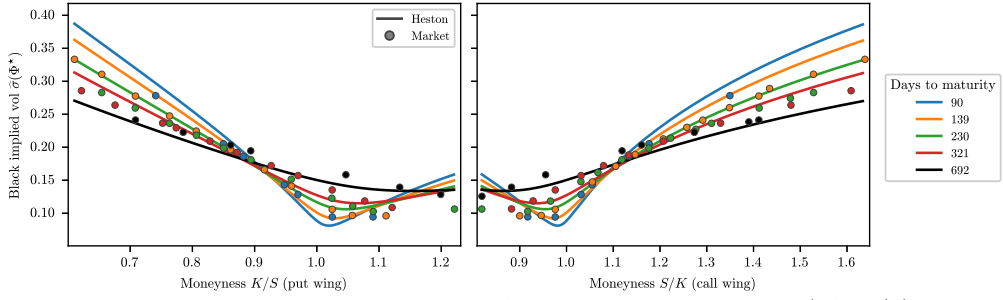


FIGURE 10. Heston implied-volatility smiles for January 27, 2017: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

2015-05-22: $S_{ref} = 2129.02$, $\mathcal{F} = -1.4704$, IV-RMSE = 0.0057, RMSE = 0.2020
 $\Phi^* = (0.0454, 3.6931, 1.3437, -0.7294, 0.0081)$

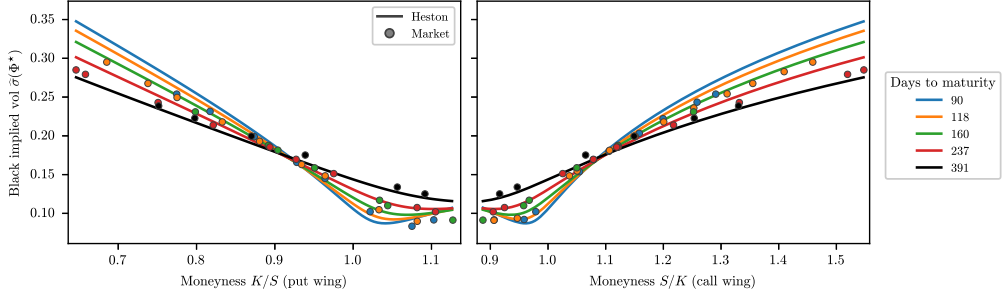


FIGURE 11. Heston implied-volatility smiles for May 22, 2015: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

2015-05-28: $S_{ref} = 2117.65$, $\mathcal{F} = -1.7266$, IV-RMSE = 0.0058, RMSE = 0.2016
 $\Phi^* = (0.0434, 4.1796, 1.4455, -0.7511, 0.0134)$

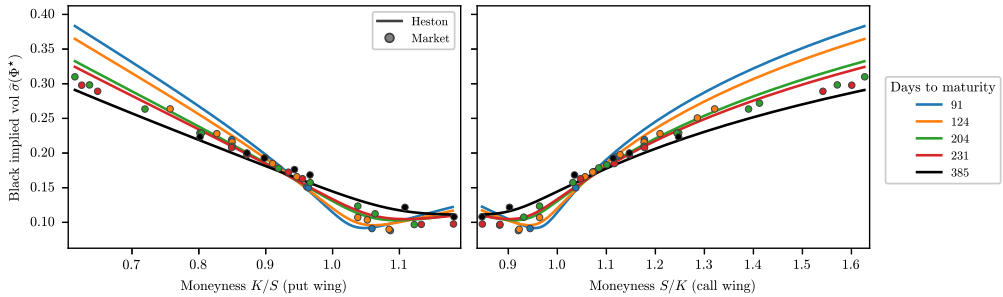


FIGURE 12. Heston implied-volatility smiles for May 28, 2015: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

Appendix C. Example fits for the Bates model

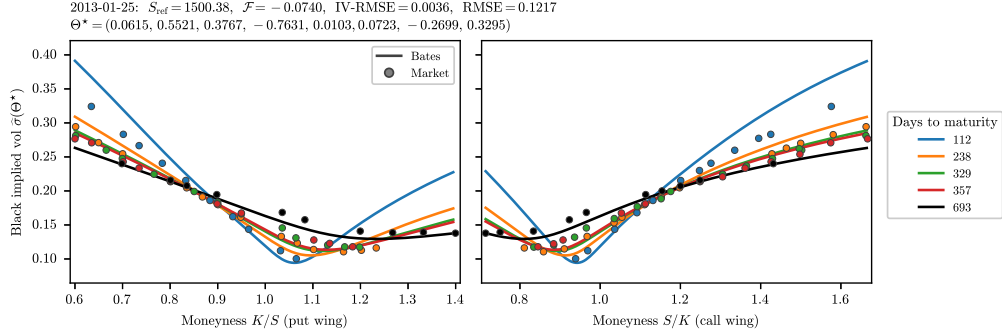


FIGURE 13. Bates implied-volatility smiles for January 25, 2013: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

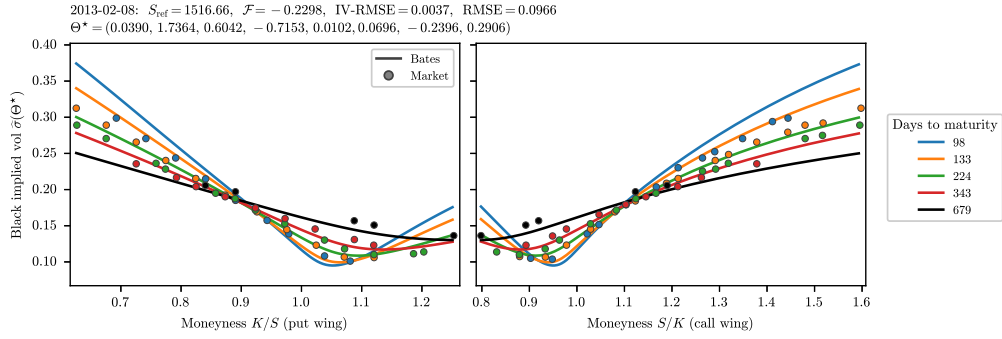


FIGURE 14. Bates implied-volatility smiles for February 08, 2013: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

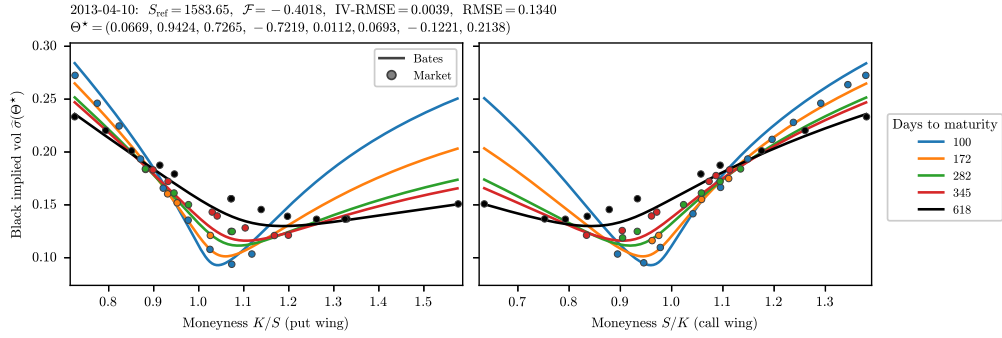


FIGURE 15. Bates implied-volatility smiles for April 10, 2013: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

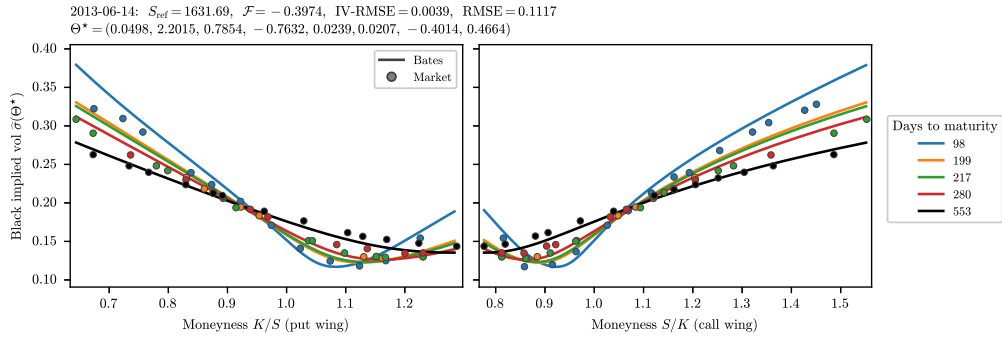


FIGURE 16. Bates implied-volatility smiles for June 14, 2013: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

2013-02-11: $S_{ref} = 1516.36$, $\mathcal{F} = -0.5837$, IV-RMSE = 0.0039, RMSE = 0.1062
 $\Theta^* = (0.0424, 2.0805, 0.8718, -0.6801, 0.0104, 0.0927, -0.1469, 0.2154)$

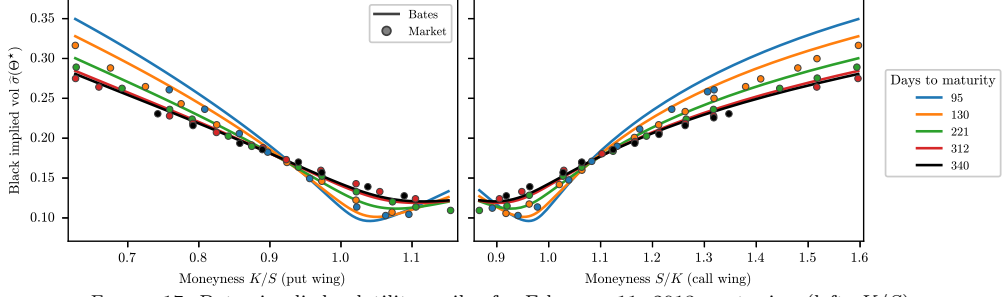


FIGURE 17. Bates implied-volatility smiles for February 11, 2013: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

2013-05-06: $S_{ref} = 1617.28$, $\mathcal{F} = -0.2013$, IV-RMSE = 0.0040, RMSE = 0.1424
 $\Theta^* = (0.0505, 1.0074, 0.5505, -0.6750, 0.0125, 0.0244, -0.3850, 0.4548)$

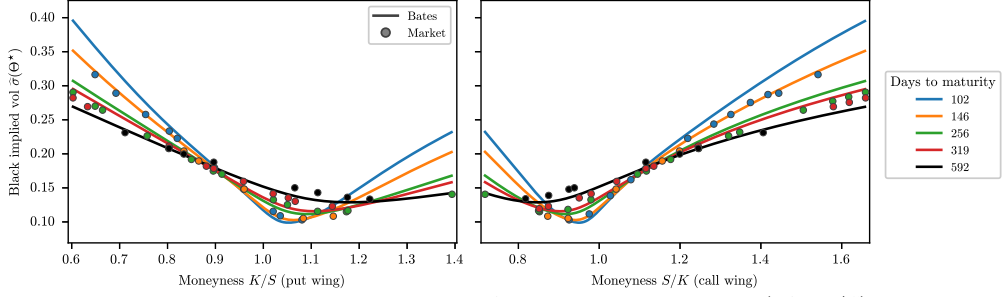


FIGURE 18. Bates implied-volatility smiles for May 06, 2013: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

2013-07-09: $S_{ref} = 1649.59$, $\mathcal{F} = -0.1190$, IV-RMSE = 0.0040, RMSE = 0.1250
 $\Theta^* = (0.0603, 0.7789, 0.4614, -0.7368, 0.0169, 0.0285, -0.3997, 0.5000)$

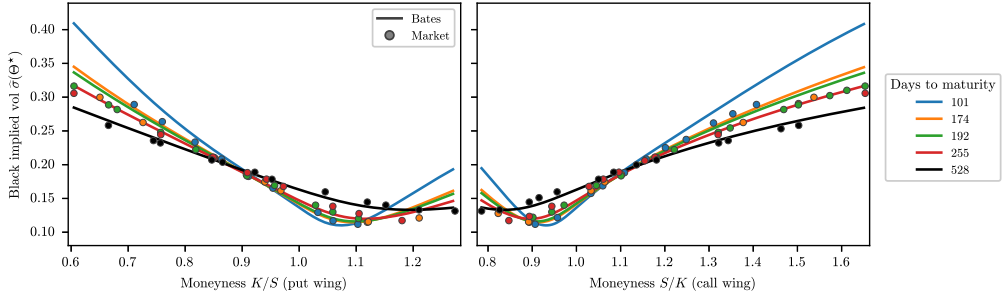


FIGURE 19. Bates implied-volatility smiles for July 09, 2013: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

2013-06-18: $S_{ref} = 1648.36$, $\mathcal{F} = -0.5224$, IV-RMSE = 0.0040, RMSE = 0.1202
 $\Theta^* = (0.0513, 1.9739, 0.8513, -0.7539, 0.0240, 0.0393, -0.2191, 0.3321)$

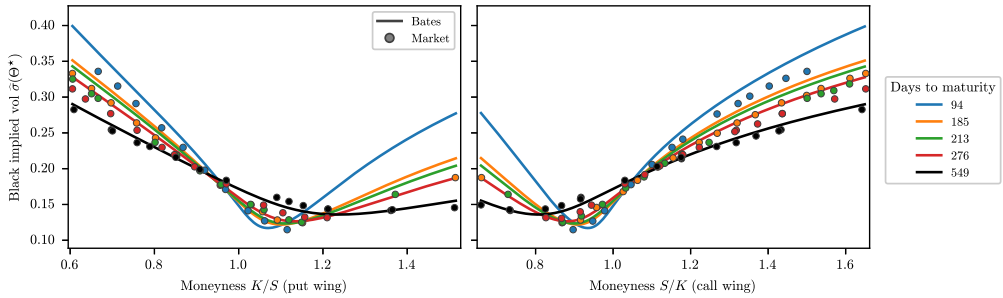


FIGURE 20. Bates implied-volatility smiles for June 18, 2013: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

Appendix D. Algorithm

Algorithm 1 states the per-day routine in the active vector Ψ ; the driver applies it to every trading day and accumulates accepted rows into a single file (§7). Heston is the default ($\Psi = \Phi$). The Bates fit is the same algorithm with $\Psi = \Theta$: the engine is the **BatesEngine** and the box $\mathcal{B}$ is the dim Θ -dimensional one, while the pegging test (35) still ranges over the dim Φ diffusive coordinates only.

Algorithm 1 Per-day calibration

Require: day d of OTM trades; rates $r(d), g(d)$; constants of Table 1

- 1: $\mathcal{D} \leftarrow \{i : \sigma_i > 0, \tau_i \geq \tau_{\min}\}$ (pre-filter (17))
- 2: **if** $r(d)$ or $g(d)$ undefined on/before d **then**
- 3: **return** $\emptyset$ (no rate)
- 4: **end if**
- 5: $S_{\text{ref}} \leftarrow (\sum q_i S_i) / (\sum q_i)$;
 $R \leftarrow \max_i S_i / \min_i S_i - 1$;
 $\text{HIGH_MOVE} \leftarrow \mathbf{1}\{R > R_{\max}\}$
- 6: $K_i^* \leftarrow \Delta \text{round}(m_i S_{\text{ref}} / \Delta)$ with $m_i = K_i / S_i$ (re-strike (20))
- 7: $\mathcal{T} \leftarrow \text{top-}\bar{N}_t$ admissible maturities by volume ($\geq \underline{n}_k$ strikes per wing);
 build $\mathcal{K}_\tau$ (maximum $\bar{n}_k$ per wing);
 pivot to $\{\hat{\sigma}_{k,\tau}\}$, cells $\mathcal{N}$
- 8: **if** coverage (25) fails **then**
- 9: clear stale diagnostics; **return** $\emptyset$
- 10: **end if**
- 11: $\hat{v} \leftarrow \text{clip}(\text{median}(\hat{\sigma})^2, [10^{-3}, 0.25])$; build seeds $\mathcal{S}$ (31) (32)
- 12: **for** $s \in \mathcal{S}$ **do**
- 13: $\hat{\Psi}_s \leftarrow \text{LM minimiser of } \sum_n (\varepsilon_n^{\text{obj}})^2 \text{ on } \mathcal{B} \text{ from } s$ (33)
- 14: compute $J_{\text{iv}}(\hat{\Psi}_s)$ (30)
- 15: **end for**
- 16: $\Psi^* \leftarrow \arg \min_s J_{\text{iv}}(\hat{\Psi}_s)$ (select (34))
- 17: **if** $J_{\text{iv}}(\Psi^*) \leq \varepsilon_{\text{iv}}$ **and** $\neg \text{PEGGED}(\Psi^*)$ **then**
- 18: write repriced diagnostics; **return** $\text{row}(d, S_{\text{ref}}, r, g, \Psi^*, \mathcal{F}, J_{\text{iv}}, J_{\text{px}}, \dots)$
- 19: **else**
- 20: clear stale diagnostics; **return** $\emptyset$ (rejected)
- 21: **end if**
